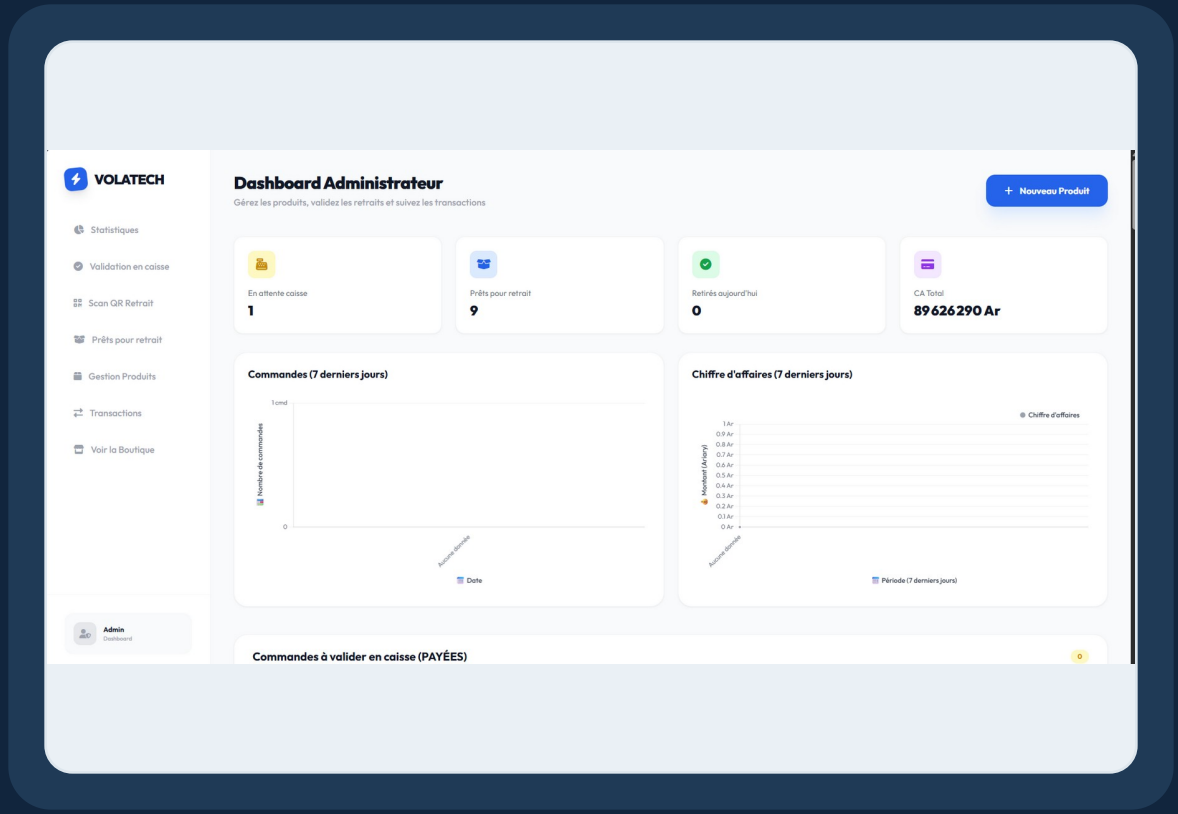
TECHNICAL CASE STUDY · ENGLISH EDITION

VOLATECH

Web-based product sales and collection platform.



JAVA EE WEB PROJECT · 2025

Computer retail workflows, checkout, invoices and QR-assisted pickup.

TECH STACK

Java 17 · Jakarta EE · JSP · Servlets · PostgreSQL

Presented by: RANDRIAMANANTENA Fitahiana Lafatra

Project overview

An e-commerce workflow with validation and pickup.

DOMAIN · COMPUTER RETAIL

Product catalog, cart, orders and purchase tracking.

PROCESS · FROM CART TO PICKUP

Order validation, invoicing and QR-based pickup workflow.

ACTORS · CUSTOMER AND ADMIN

Separate interfaces for shopping and store operations.

ARCHITECTURE · TRADITIONAL JAVA WEB

Servlets/JSP, Java services, JDBC data access and PostgreSQL.

A coherent purchase workflow

Connecting business functions that involve several workflow states.

01 · CATALOG

Display categories, prices and product availability.

02 · ORDER

Keep the shopping cart, order lines, amounts and status.

03 · PICKUP

Create the invoice, then track validation and product pickup.

IMPORTANT · The documented payment mode is simulated, not actual bank collection.

From catalog to pickup

Process designed for a store with controlled product collection.

01 · SHOP

Browse the catalog and build a cart.

02 · ORDER

Record the order and its items.

03 · VALIDATE

Process a simulated payment step, followed by checkout validation.

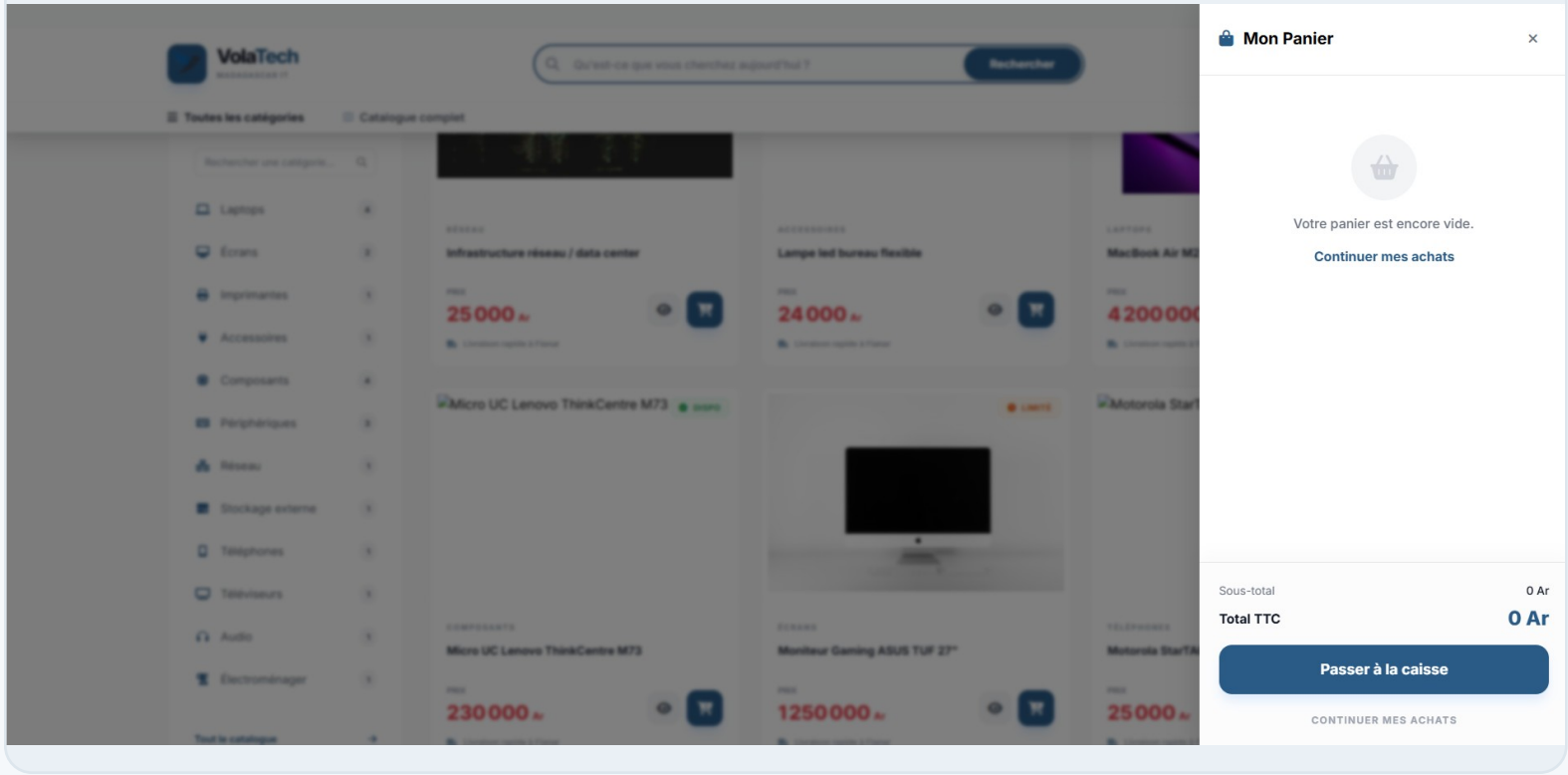
04 · COLLECT

Present the QR code and update the pickup status.

REMEMBER · Concurrency controls and complete workflow execution still require full testing.

Storefront and side cart

Public purchasing journey.



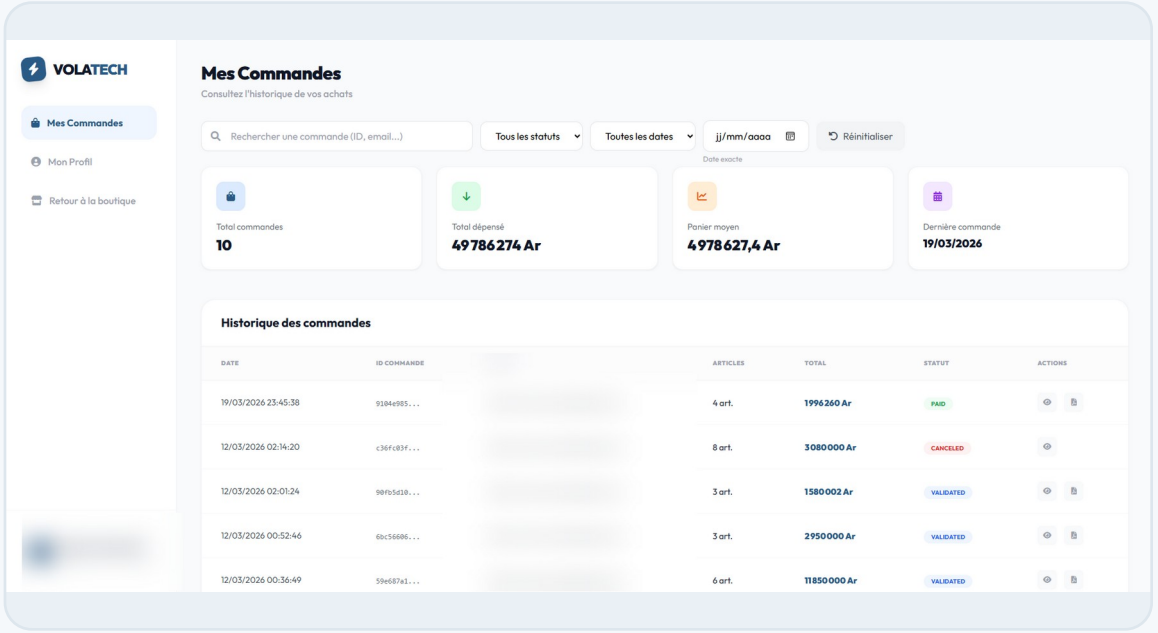
Interface screenshot; products and prices are used for demonstration.

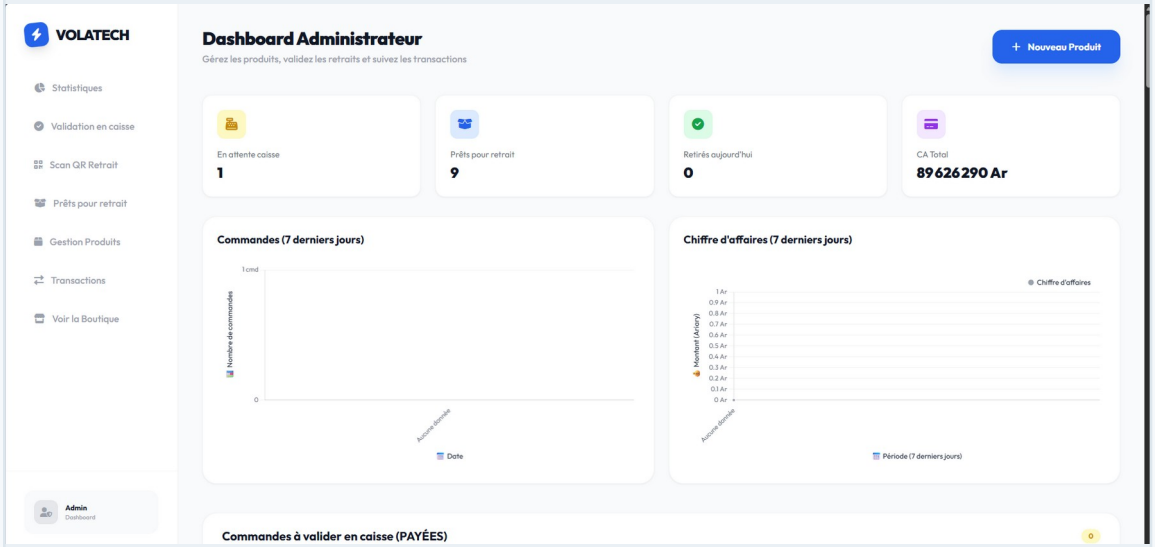
WHAT THE VIEW SHOWS

Categorized catalog of computer products; search and price consultation; cart creation; access to order confirmation.

Customer space and Admin dashboard

Order tracking and activity monitoring.





CUSTOMER AREA

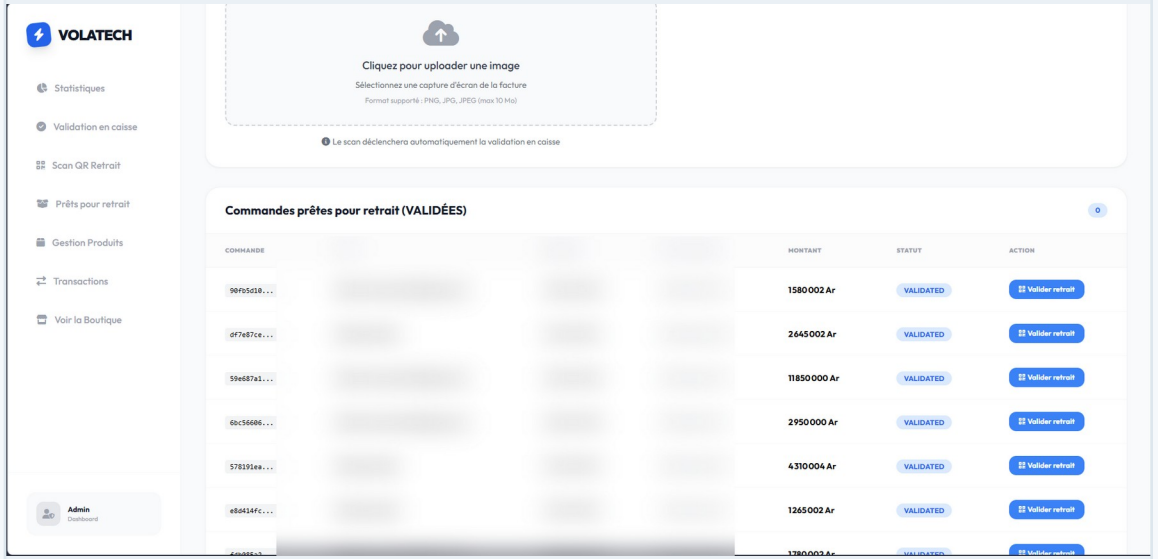
Order history, filters, statuses and totals.

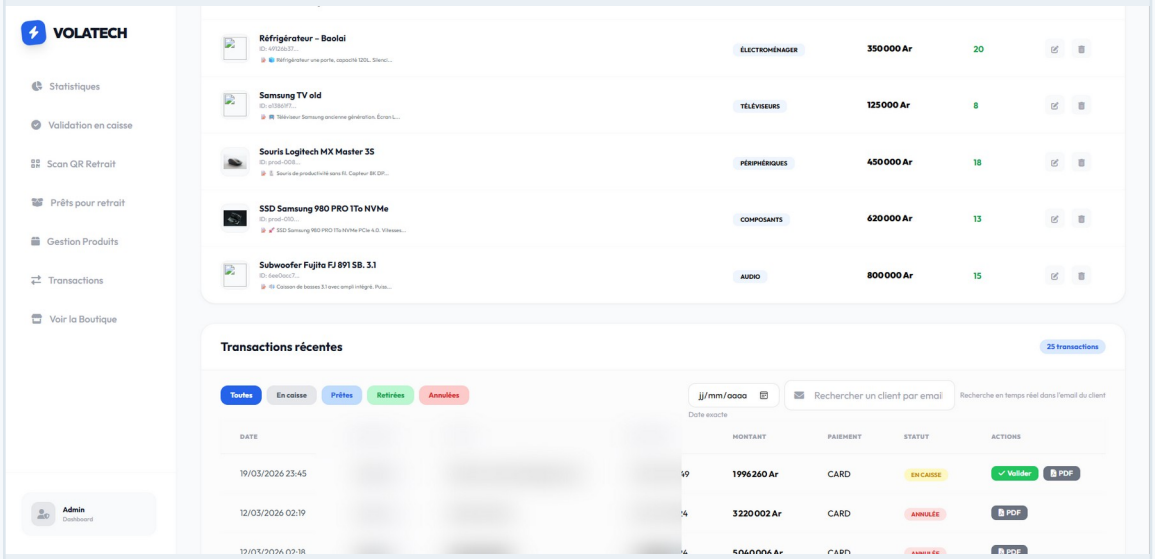
ADMIN MANAGEMENT

Order KPIs and chart placeholders for activity monitoring.

Checkout and transaction management

Cashier validation and stock/transaction oversight.





CHECKOUT VALIDATION

Validate orders at checkout and prepare product pickup.

TRANSACTION MANAGEMENT

Track products, stocks and transaction statuses.

A layered Jakarta EE web application

WAR packaging for a Jakarta EE server, with WildFly configuration planned.

01 · USER INTERFACE

JSP / JavaScript. HTML/CSS, Tailwind and Chart.js.

02 · HTTP

Servlets. Requests, sessions and role checks.

03 · BUSINESS LOGIC

Java services. Orders, simulated payment, checkout and invoices.

04 · SQL

JDBC. Repositories and PostgreSQL via DataSource.

CROSS-CUTTING · JSP/Servlets and JDBC define this project; it is not a Spring Boot application.

Seven PostgreSQL tables

Key entities and business relationships in the documented model.

USERS

Identities and roles.

CATEGORIES

Product groups.

PRODUCTS

Items, prices and stock.

ORDERS

Orders and statuses.

ORDER_ITEMS

Order lines, quantities and prices.

PAYMENTS / INVOICES

Payments and invoices.

RELATIONSHIPS · users → orders → order_items → products → categories; orders → payments / invoices. Seven tables are identified in the archive; row counts and complete execution are not claimed.

Order and status management

Status transitions are business rules and must be secured on the server.

01 · ORDER

Build the order lines and validate the items.

02 · SIMULATION

Produce an internal payment reference such as SIM-… .

03 · CHECKOUT

Apply an administrative validation.

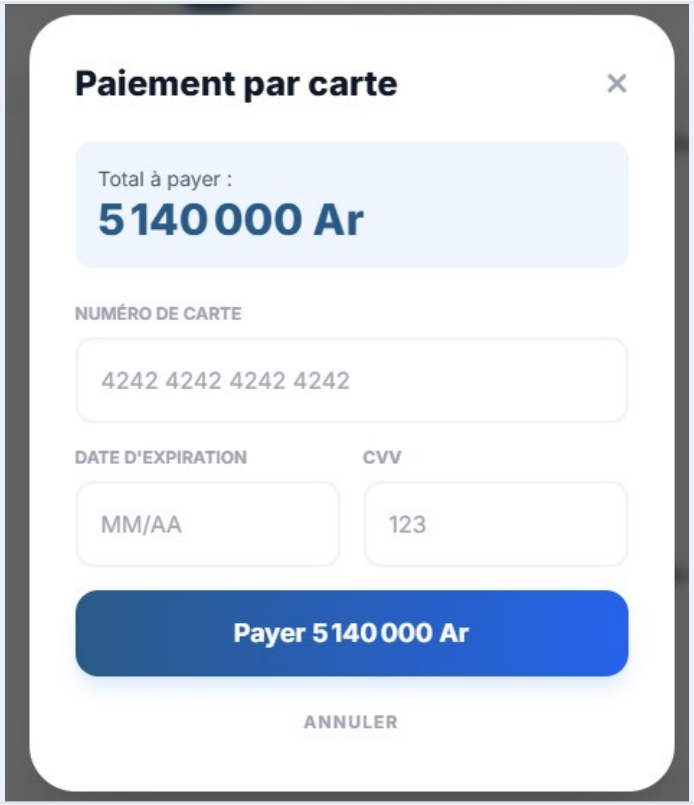
04 · PICKUP

Check the QR code and record product collection.

REMEMBER · Application statuses do not constitute confirmation of actual bank payments.

Card payment: simulated interface

Illustrated checkout with an amount in ariary; no bank debit is demonstrated.



Supplied screenshot · test form with example card/CVV.

WHAT THE SCREENSHOT SHOWS

Total amount, example card/expiry/CVV fields and payment confirmation.

PROJECT STATUS

Fictional card payment scenario; internal transaction reference may start with SIM-… .

CORRECT INTERPRETATION

A UI illustrating a payment flow, not evidence of a debit, a payment service provider or PCI compliance.

From simulation to a real bank integration

The existing interface could serve as a starting point, but sensitive entry must be replaced with a secure provider component.

01 · CREATE THE ORDER

The server calculates the amount and creates a pending payment.

02 · USE A PSP

Redirect or use hosted fields from an authorized provider.

03 · AUTHENTICATE

The provider handles the card; 3-D Secure if applicable.

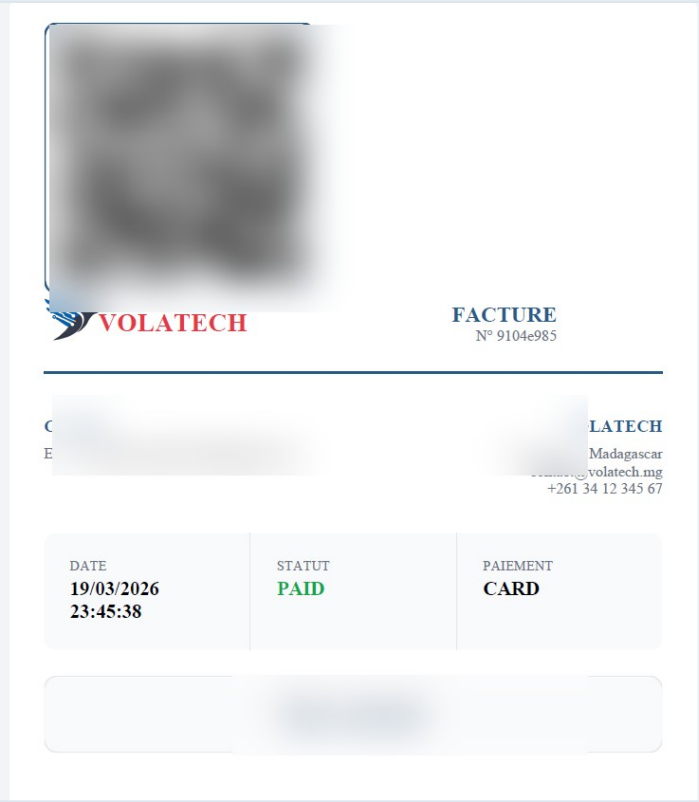
04 · CONFIRM SERVER-SIDE

Signed webhook / API verification, idempotency and final status.

SECURITY RULE · VolaTech must not store or transmit card numbers or CVVs. Sensitive collection belongs to the payment provider. This is a future direction, not implemented or validated in the version shown.

PDF invoice and pickup QR code

Supporting documents within the order workflow.


VOLATECH

FACTURE

N° 9104e985

LATECH

Madagascar

volatech.mg

+261 34 12 345 67

DATE

19/03/2026

23:45:38

STATUT

PAID

PAIEMENT

CARD

INVOICING

PDF generated by the Java server with an order reference.

QR CODE

Reference associated with collection, for use in the verification interface.

CAUTION

The QR is not presented as cryptographically signed; illustrative use only.

Technical mechanisms and limitations

Screenshots establish the interfaces, not production readiness.

STRUCTURE · LAYER SEPARATION

Servlets, services, repositories and parameterized SQL access.

ACCESS · SESSIONS AND AUTHORIZATION

Customer/Admin spaces; private data checks still need strengthening.

ROBUSTNESS · TRANSACTIONS AND STOCK

Atomic operations, concurrency, idempotency and end-to-end tests need to be consolidated.

SCOPE · Demonstration project: simulated financial transactions and no proof of production use.

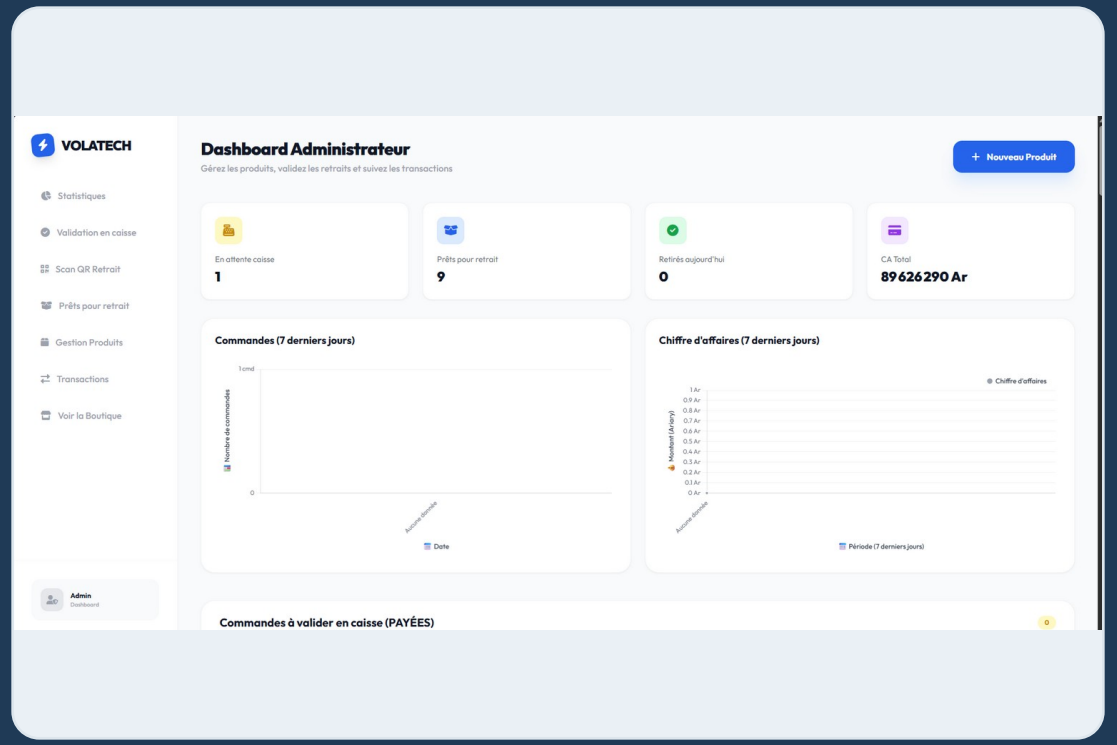
TECHNICAL CASE STUDY · ENGLISH EDITION

VOLATECH

A Java EE case study linking e-commerce journeys, order logic and PostgreSQL persistence.

KEY THEMES

Servlets & JSP · SQL / JDBC · PDF invoicing · Pickup QR



Screenshots and architecture based on project documentation. No production deployment is assumed.

Presented by: RANDRIAMANANTENA Fitahiana Lafatra